Parul University			
Academic Year: 2026-2027			
Name of Teacher: ASHWANI CHAUDHARY			
Subject: Artificial Intelligence Laboratory			
Subject Code:303105308			
Lesson/Lab Plan No	**Lesson/Lab Plan Title**	**Lesson/Lab Plan Description**	**Lesson/Lab Plan Date (dd-mm-yyyy)**
1	Practical - 1	Develop an AI-based medical diagnosis system using expert systems architecture and knowledge representation techniques.	6/12/2026
2	Practical - 2	Implement a constraint satisfaction algorithm to solve scheduling problems in healthcare facilities.	6/19/2026
3	Practical - 3	Develop a problem-solving agent for optimizing resource allocation in logistics using A* and AO* algorithms.	6/26/2026
4	Practical - 4	Write a program to implement BFS (Water Jug problem or any AI search problem).	7/3/2026
5	Practical - 5	Define a predicate brother(X,Y) which holds iff X and Y are brothers. Define a predicate cousin(X,Y) which holds iff X and Y are cousins. Define a predicate grandson(X,Y) which holds iff X is a grandson of Y. Define a predicate descendent(X,Y) which holds iff X is a descendent of Y. Consider the following genealogical tree: father(a,b). father(a,c). father(b,d). father(b,e). father(c,f). Say which answers, and in which order, are generated by your definitions for the following queries in Prolog: ?- brother(X,Y). ?- cousin(X,Y). ?- grandson(X,Y). ?- descendent(X,Y).	7/10/2026
6	Practical - 6	Write a program to implement Tic-Tac-Toe game using python.	7/17/2026
7	Practical - 7	Create a spell-checking application utilizing natural language processing (NLP) techniques, including syntactic and semantic	7/24/2026
8	Practical - 8	Design a neural network architecture for pattern recognition in medical imaging for disease diagnosis.	7/31/2026
9	Practical - 9	Build an intelligent agent for optimizing e-commerce inventory management using search algorithms like hillclimbing and best-first search.	8/7/2026
10	Practical - 10	Create a recommendation system for personalized learning using means-end analysis and heuristic search techniques.	8/14/2026
11	Practical - 11	Develop a fuzzy logic-based system for predicting stock market trends considering uncertain market conditions..	9/11/2026
12	Practical - 12	Write a program to implement DFS (Water Jug problem or any AI search problem)	9/18/2026
13	Practical - 13	Family Tree Representation using Prolog	9/18/2026
14	Practical - 14	Implementation of Neural Network concepts	9/25/2026
15	Practical - 15	Knowledge Representation using Predicate Logic	9/25/2026